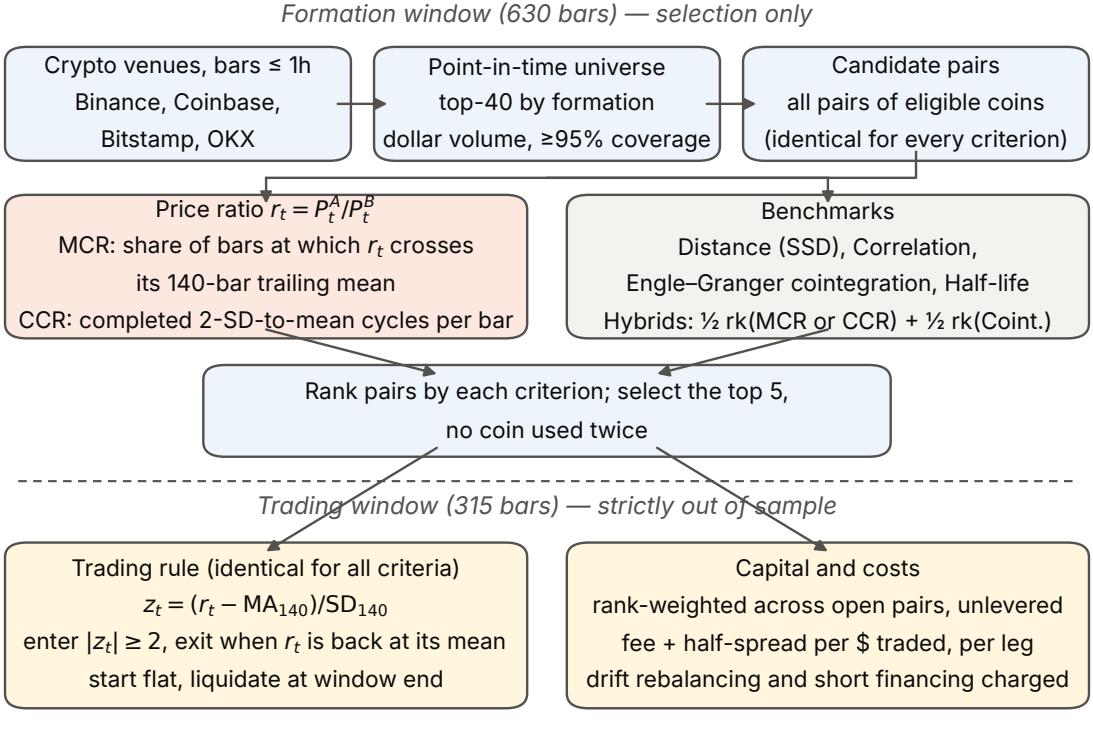
Formation window (630 bars) — selection only
Crypto venues, bars ≤ 1h
Binance, Coinbase,
Bitstamp, OKX
Point-in-time universe
top-40 by formation
dollar volume, ≥95% coverage
Candidate pairs
all pairs of eligible coins
(identical for every criterion)
Price ratio $r_t = P_t^A / P_t^B$
MCR: share of bars at which r_t crosses
its 140-bar trailing mean
CCR: completed 2-SD-to-mean cycles per bar
Benchmarks
Distance (SSD), Correlation,
Engle–Granger cointegration, Half-life
Hybrids: ½ rk(MCR or CCR) + ½ rk(Coint.)
Rank pairs by each criterion; select the top 5,
no coin used twice
Trading window (315 bars) — strictly out of sample
Trading rule (identical for all criteria)
$z_t = (r_t - \mathrm{MA}_{140})/\mathrm{SD}_{140}$
enter $|z_t| \geq 2$, exit when r_t is back at its mean
start flat, liquidate at window end
Capital and costs
rank-weighted across open pairs, unlevered
fee + half-spread per $ traded, per leg
drift rebalancing and short financing charged